

Measurements, spectral decomposition and density matrices

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Summary from last week and plans for this week

Last week we:

1. defined the state vector and the associated notation
2. introduced the inner product and showed how to calculate it in an orthonormal basis
3. introduced outer products and projection operators
4. introduced tensor products and showed how to construct state vectors for multiple qubits

This week's plans

We will repeat some of the topics from last week and discuss

1. tensor products of Hilbert Spaces and definition of Computational basis, partly repetition from last week
2. Bloch sphere and representation of qubits
3. Spectral Decomposition, Measurements and Density matrices
4. Wavefunction collapse as a result of measurement
5. Entanglement and relations to density matrices

Readings

- ▶ **Reading recommendation:** Scherer, Mathematics of Quantum Computations, parts of chapter 2 and sections 3.1-3.3 and Hundt, Quantum Computing for Programmers, chapter 2.1-2.5. Hundt's text is relevant for the programming part where we build from scratch the ingredients we will need.

Feel free to study the exercises in Scherer's text as well. Solutions are available.

A good read is also Audtresh's text on **Entangled systems**. For density matrices see chapter 4 at <https://github.com/CompPhysics/QuantumComputingMachineLearning/blob/gh-pages/doc/Textbooks/EntanglementBook.pdf>

Zoom link

Reminder about the zoom link at
<https://uio.zoom.us/my/mortenhj>

Measurements

The probability of a measurement on a quantum system giving a certain result is determined by the weight of the relevant basis state in the state vector. After the measurement, the system is in a state that corresponds to the result of the measurement. The operators and gates discussed below are examples of operations we can perform on specific states.

We consider the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Properties of a measurement

1. A measurement can yield only one of the above states, either $|0\rangle$ or $|1\rangle$.
2. The probability of a measurement resulting in $|0\rangle$ is $\alpha^* \alpha = |\alpha|^2$.
3. The probability of a measurement resulting in $|1\rangle$ is $\beta^* \beta = |\beta|^2$.
4. And we note that the sum of the outcomes gives $\alpha^* \alpha + \beta^* \beta = 1$ since the two states are normalized.

After the measurement, the state of the system is the state associated with the result of the measurement.

We have already encountered the projection operators P and Q . Let us now look at other types of operations we can make on qubit states.

Basic properties of hermitian operators

The operators we typically encounter in quantum mechanical studies are

1. Hermitian (self-adjoint) meaning that for example the elements of a Hermitian matrix $\mathbf{U}$ obey $u_{ij} = u_{ji}^*$.
2. Unitary $\mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U} = \mathbf{I}$, where $\mathbf{I}$ is the unit matrix
3. The operator $\mathbf{U}$ and its self-adjoint commute (often labeled as normal operators), that is $[\mathbf{U}, \mathbf{U}^\dagger] = 0$. An operator is **normal** if and only if it is diagonalizable. A Hermitian operator is normal.

Unitary operators in a Hilbert space preserve the norm and orthogonality. If $\mathbf{U}$ is a unitary operator acting on a state $|\psi_j\rangle$, the action of

$$|\phi_i\rangle = \mathbf{U}|\psi_j\rangle,$$

preserves both the norm and orthogonality, that is $\langle\phi_i|\phi_j\rangle = \langle\psi_i|\psi_j\rangle = \delta_{ij}$, as discussed earlier.

Spectral Decomposition

An important technicality which we will use in the discussion of density matrices, entanglement, quantum entropies and other properties is the so-called spectral decomposition of an operator. Let $|\psi\rangle$ be a vector in a Hilbert space of dimension n and a hermitian operator $\mathbf{A}$ defined in this space. Assume $|\psi\rangle$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ , that is

$$\mathbf{A}|\psi\rangle = \lambda|\psi\rangle = \lambda\mathbf{I}|\psi\rangle,$$

where we used $\mathbf{I}|\psi\rangle = 1|\psi\rangle$. Subtracting the right hand side from the left hand side gives

$$[\mathbf{A} - \lambda\mathbf{I}]|\psi\rangle = 0,$$

which has a nontrivial solution only if the determinant $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.

ONB again and again

We define now an orthonormal basis $|i\rangle = \{|0\rangle, |1\rangle, \dots, |n-1\rangle$ in the same Hilbert space. We will assume that this basis is an eigenbasis of $\mathbf{A}$ with eigenvalues λ_i

We expand a new vector using this eigenbasis of $\mathbf{A}$

$$|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i |i\rangle,$$

with the normalization condition $\sum_{i=0}^{n-1} |\alpha_i|^2 = 1$. Acting with $\mathbf{A}$ on this new state results in

$$\mathbf{A}|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i \mathbf{A}|i\rangle = \sum_{i=0}^{n-1} \alpha_i \lambda_i |i\rangle.$$

Projection operators

If we then use that the outer product of any state with itself defines a projection operator we have the projection operators

$$\mathbf{P}_\psi = |\psi\rangle\langle\psi|,$$

and

$$\mathbf{P}_j = |j\rangle\langle j|,$$

we have that

$$\mathbf{P}_j|\psi\rangle = |j\rangle\langle j| \sum_{i=0}^{n-1} \alpha_i |i\rangle = \sum_{i=0}^{n-1} \alpha_i |j\rangle\langle j|i\rangle.$$

Further manipulations

This results in

$$\mathbf{P}_j|\psi\rangle = \alpha_j|j\rangle,$$

since $\langle j|i\rangle = \delta_{ji}$. With the last equation we can rewrite

$$\mathbf{A}|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i \lambda_i |i\rangle = \sum_{i=0}^{n-1} \lambda_i \mathbf{P}_i |\psi\rangle,$$

from which we conclude that

$$\mathbf{A} = \sum_{i=0}^{n-1} \lambda_i \mathbf{P}_i.$$

Spectral decomposition

This is the spectral decomposition of a hermitian and normal operator. It is true for any state and it is independent of the basis. The spectral decomposition can in turn be used to exhaustively specify a measurement, as we will see in the next section.

As an example, consider two states $|\psi_a\rangle$ and $|\psi_b\rangle$ that are eigenstates of $\mathbf{A}$ with eigenvalues λ_a and λ_b , respectively. In the diagonalization process we have obtained the coefficients α_0 , α_1 , β_0 and β_1 using an expansion in terms of the orthogonal basis $|0\rangle$ and $|1\rangle$.

Explicit results

We have then

$$|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle,$$

and

$$|\psi_b\rangle = \beta_0|0\rangle + \beta_1|1\rangle,$$

with corresponding projection operators

$$\mathbf{P}_a = |\psi_a\rangle\langle\psi_a| = \begin{bmatrix} |\alpha_0|^2 & \alpha_0\alpha_1^* \\ \alpha_1\alpha_0^* & |\alpha_1|^2 \end{bmatrix},$$

and

$$\mathbf{P}_b = |\psi_b\rangle\langle\psi_b| = \begin{bmatrix} |\beta_0|^2 & \beta_0\beta_1^* \\ \beta_1\beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

The spectral decomposition

The results from the previous slide gives us the following spectral decomposition of $\mathbf{A}$

$$\mathbf{A} = \lambda_a |\psi_a\rangle \langle \psi_a| + \lambda_b |\psi_b\rangle \langle \psi_b|,$$

which written out in all its details reads

$$\mathbf{A} = \lambda_a \begin{bmatrix} |\alpha_0|^2 & \alpha_0 \alpha_1^* \\ \alpha_1 \alpha_0^* & |\alpha_1|^2 \end{bmatrix} + \lambda_b \begin{bmatrix} |\beta_0|^2 & \beta_0 \beta_1^* \\ \beta_1 \beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

Bloch sphere

Classically, in a binary system, the bits take only two distinct values, either 0 or 1. The quantum mechanical counterpart is given by two state vectors (our simple computational basis) $|0\rangle$ and $|1\rangle$ which can be used to realize the superposition

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

which can be represented using the so-called Bloch sphere, depicted on the next slide (best seen using the jupyter-notebook).

Meet the Bloch sphere

The Bloch sphere gives a viable way to visualize a qubit and its possible realizations in terms of the angles $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$.

```
import numpy as np
from qiskit.visualization import plot_bloch_vector
plot_bloch_vector([0,1,0], title="New Bloch Sphere")
```

You can use spherical coordinates instead of cartesian ones.

```
plot_bloch_vector([1, np.pi/2, np.pi/3], coord_type='spherical')
```

Using the Bloch sphere representation of the qubit

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we can rewrite it as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)\exp(i\phi)|1\rangle,$$

Bloch sphere exercise

Determine the Bloch sphere angles θ and ϕ for the eigenstates of each Pauli matrix (see lectures from last week for the definition of Pauli matrices).

Measurements

Armed with the spectral decomposition, we are now ready to discuss how to compute measurements of observables. When we make a measurement, quantum mechanics postulates that mutually exclusive measurement outcomes correspond to orthogonal projection operators.

We assume now we can construct a series of such orthogonal operators based on $|i\rangle \in \{|0\rangle, |1\rangle, \dots, |n-1\rangle$ computational basis states. These projection operators $P_0, P_1, \dots, P_{n-1}$ are all idempotent and sum to one

$$\sum_{i=0}^{n-1} P_i = I.$$

Qubit example

As an example, consider the basis of two qubits $|0\rangle$ and $|1\rangle$ with the corresponding sum

$$\sum_{i=0}^1 \mathbf{P}_i = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Based on the spectral decomposition discussed above, we can define the probability of eigenvalue λ_i as

$$\text{Prob}(\lambda_i) = |\mathbf{P}_i|\psi\rangle|^2,$$

where $|\psi_a\rangle$ is a quantum state representing the system prior to a specific measurement.

Total probability

We can rewrite this as

$$\text{Prob}(\lambda_i) = \langle \psi | \mathbf{P}_i^\dagger \mathbf{P}_i | \psi \rangle = \langle \psi | \mathbf{P}_i | \psi \rangle.$$

The total probability for all measurements is the sum over all probabilities

$$\sum_{i=0}^{n-1} \text{Prob}(\lambda_i) = 1.$$

We can in turn define the post-measurement normalized pure quantum state as, for the specific outcome λ_i , as

$$|\psi'\rangle = \frac{\mathbf{P}_i |\psi\rangle}{\sqrt{\langle \psi | \mathbf{P}_i | \psi \rangle}}.$$

Binary example system

As an example, consider the binary system states $|0\rangle$ and $|1\rangle$ with corresponding projection operators

$$P_0 = |0\rangle\langle 0|,$$

and

$$P_1 = |1\rangle\langle 1|,$$

with the properties

$$\sum_{i=0}^1 P_i^\dagger P_i = I,$$

$$P_0^\dagger P_0 = P_0^2 = P_0,$$

and

$$P_1^\dagger P_1 = P_1^2 = P_1.$$

Superposition of states

Assume thereafter that we have a state $|\psi\rangle$ which is a superposition of the above two qubit states

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

The probability of finding either $|0\rangle$ or $|1\rangle$ is then

$$\mathbf{P}_{\psi(0)} = \langle\psi|\mathbf{P}_0^\dagger\mathbf{P}_0|\psi\rangle = |\alpha|^2,$$

and similarly we have

$$\mathbf{P}_{\psi(1)} = \langle\psi|\mathbf{P}_1^\dagger\mathbf{P}_1|\psi\rangle = |\beta|^2.$$

More derivations

If we set

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle),$$

we have $|\alpha|^2 = |\beta|^2 = 1/2$. In general for this system we have

$$|\psi'_0\rangle = \frac{\mathbf{P}_0|\psi\rangle}{\sqrt{\langle\psi|\mathbf{P}_0|\psi\rangle}} = \frac{\alpha}{|\alpha|}|0\rangle,$$

and

$$|\psi'_1\rangle = \frac{\mathbf{P}_1|\psi\rangle}{\sqrt{\langle\psi|\mathbf{P}_1|\psi\rangle}} = \frac{\beta}{|\beta|}|1\rangle.$$

Final result

In general we have that

$$P_{\psi(x)} = \langle \psi | P_x^\dagger P_x | \psi \rangle, ,$$

which we can rewrite as

$$\text{Prob}(\psi(x)) = \text{Tr} \left[P_x^\dagger P_x | \psi \rangle \langle \psi | \right] .$$

Example

The last equation can be understood better through the following example with a state $|\psi\rangle$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

which results in a projection operator

$$|\psi\rangle\langle\psi| = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}.$$

Computing matrix products

We have that

$$\mathbf{P}_0^\dagger \mathbf{P}_0 = \mathbf{P}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

and computing the matrix product $\mathbf{P}_0 |\psi\rangle\langle\psi|$ gives

$$\mathbf{P}_0 |\psi\rangle\langle\psi| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ 0 & 0 \end{bmatrix}.$$

Taking the trace

Taking the trace of the above matrix, that is computing

$$\text{Prob}(\psi(0)) = \text{Tr} \left[\mathbf{P}_0^\dagger \mathbf{P}_0 |\psi\rangle \langle \psi| \right] = |\alpha|^2,$$

we obtain the same results as the one we had earlier by computing the probability for 0 given by the expression

$$\mathbf{P}_{\psi(0)} = \langle \psi | \mathbf{P}_0^\dagger \mathbf{P}_0 | \psi \rangle = |\alpha|^2.$$

Outcome probability

It is straight forward to show that

$$\text{Prob}(\psi(1)) = \text{Tr} \left[\mathbf{P}_1^\dagger \mathbf{P}_1 |\psi\rangle\langle\psi| \right] = |\beta|^2,$$

which we also could have obtained by computing

$$\mathbf{P}_{\psi(1)} = \langle\psi| \mathbf{P}_1^\dagger \mathbf{P}_1 |\psi\rangle = |\beta|^2.$$

Extending the expressions

We can now extend these expressions to the complete ensemble of measurements. Using the spectral decomposition we have that the probability of an outcome $p(x)$ is

$$p(x) = \sum_{i=0}^{n-1} p_i \mathbf{P}_{\psi_i(x)},$$

where p_i are the probabilities of a specific outcome.

With these prerequisites we are now ready to introduce the density matrices, or density operators.

Density Matrices

As opposed to the wavefunction (a so-called pure state) in closed (Hermitian) quantum mechanics, the density matrix uniquely describes the state of a quantum system in an open system, as well as in closed systems.

With pure states, the state of a quantum system is deterministic and leads to probabilities of outcomes of measurements. In contrast, what if the state itself is uncertain and therefore probabilistic? This is what density matrices describe.

Density matrices/operators

The last equation can be rewritten as

$$p(x) = \sum_{i=0}^{n-1} p_i \mathbf{P}_{\psi_i(x)} = \sum_{i=0}^{n-1} p_i \text{Tr} \left[\mathbf{P}_x^\dagger \mathbf{P}_x |\psi_i\rangle \langle \psi_i| \right],$$

and we define the **density matrix/operator** as

$$\rho = \sum_{i=0}^{n-1} p_i |\psi_i\rangle \langle \psi_i|,$$

we can rewrite the first equation above as

$$p(x) = \text{Tr} \left[\mathbf{P}_x^\dagger \mathbf{P}_x \rho \right].$$

If we can define the state of a system in terms of the density matrix, the probability of a specific outcome is then given by

$$p(x)_\rho = \text{Tr} \left[\mathbf{P}_x^\dagger \mathbf{P}_x \rho \right].$$

Properties of density matrices I

A density matrix in a Hilbert space with n states has the following properties (which we state without proof)

1. There exists a probability $p_i \geq 0$ with $\sum_i p_i = 1$,
2. There exists an orthonormal basis ψ_i such that we can define $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$,
3. We have $0 \leq \rho^2 \leq 1$ and
4. The norm $\|\rho\|_2 \leq 1$.

With the density matrix we can also define the state the system collapses to after a measurement, namely

$$\rho'_x = \frac{\mathbf{P}_x \rho \mathbf{P}_x^\dagger}{\text{Tr}[\mathbf{P}_x^\dagger \mathbf{P}_x \rho]}.$$

Properties of density matrices II

The density matrix of a pure state $|\psi\rangle$ is $\rho = |\psi\rangle\langle\psi|$.

ρ is Hermitian, positive semi-definite (all eigenvalues non-negative), and has unit trace

$$\rho = \rho^\dagger, \quad \lambda(\rho) \geq 0, \quad \text{Tr}(\rho) = 1.$$

Given a probabilistic mixture of states with probabilities $\{p_i\}$ and states $\{|\psi_i\rangle\}$,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|.$$

More on the density matrix III

For a $d \times d$ density matrix ρ , there exists an orthonormal basis $\{|b_i\rangle\}$ such that

$$\rho = \sum_i \lambda_i |b_i\rangle \langle b_i|,$$

where λ_i are probabilities.

The expectation value of an observable X is

$$\langle X \rangle = \text{Tr}(\rho X) = \sum_i \lambda_i \langle b_i | X | b_i \rangle.$$

The time evolution in a closed system with Hamiltonian H is

$$\rho(t) = \exp -iHt \rho(0) \exp iHt.$$

First entanglement encounter, two qubit system

We define a system that can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \quad |1\rangle_{A,B} = \begin{bmatrix} 0 & 1 \end{bmatrix}^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry.

Computational basis

This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = [1 \ 0 \ 0 \ 0]^T,$$

and

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = [0 \ 1 \ 0 \ 0]^T,$$

and

$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = [0 \ 0 \ 1 \ 0]^T,$$

and finally

$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = [0 \ 0 \ 0 \ 1]^T.$$

Bell states

The above computational basis states, which define an ONB, can in turn be used to define another ONB. As an example, consider the so-called Bell states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} [|00\rangle - |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix},$$

The next two

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

It is easy to convince oneself that these states also form an orthonormal basis.

Measurement

Measuring one of the qubits of one of the above Bell states, automatically determines, as we will see below, the state of the second qubit. To convince ourselves about this, let us assume we perform a measurement on the qubit in system A by introducing the projections with outcomes 0 or 1 as

$$P_0 = |0\rangle\langle 0|_A \otimes I_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

for the projection of the $|0\rangle$ state in system A and similarly

$$P_1 = |1\rangle\langle 1|_A \otimes I_B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

for the projection of the $|1\rangle$ state in system A .

Probability of outcome

We can then calculate the probability for the various outcomes by computing for example the probability for measuring qubit 0

$$\langle \Phi^+ | \mathbf{P}_0 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

Similarly, we obtain

$$\langle \Phi^+ | \mathbf{P}_1 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

States after measurement

After the above measurements the system is in the states

$$|\Phi'_0\rangle = \sqrt{2} [|0\rangle\langle 0|_A \otimes I_B] |\Phi^+\rangle = |00\rangle,$$

and

$$|\Phi'_1\rangle = \sqrt{2} [|1\rangle\langle 1|_A \otimes I_B] |\Phi^+\rangle = |11\rangle.$$

We see from the last two equations that the state of the second qubit is determined even though the measurement has only taken place locally on system A .

Other states

If we on the other hand consider a state like

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B,$$

this is a pure **product** state of the single-qubit, or single-particle states, of two qubits (particles) in system A and system B , respectively. We call such a state for a **pure product state**. Quantum states that cannot be written as a mixture of other states are called pure quantum states or just product states, while all other states are called mixed quantum states.

More on Bell states

A state like one of the Bell states (where we introduce the subscript AB to indicate that the state is composed of single states from two subsystem)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle_{AB} + |11\rangle_{AB}],$$

is on the other hand a mixed state and we cannot determine whether system A is in a state 0 or 1. The above state is a superposition of the states $|00\rangle_{AB}$ and $|11\rangle_{AB}$ and it is not possible to determine individual states of systems A and B , respectively.

Entanglement

We say that the state is entangled. This yields the following definition of entangled states: a pure bipartite state $|\psi\rangle_{AB}$ is entangled if it cannot be written as a product state $|\psi\rangle_A \otimes |\phi\rangle_B$ for any choice of the states $|\psi\rangle_A$ and $|\phi\rangle_B$. Otherwise we say the state is separable.

Examples of entanglement

As an example, consider an ansatz for the ground state of the helium atom with two electrons in the lowest $1s$ state (hydrogen-like orbits) and with spin $s = 1/2$ and spin projections $m_s = -1/2$ and $m_s = 1/2$. The two single-particle states are given by the tensor products of their spatial $1s$ single-particle states $|\phi_{1s}\rangle$ and their spin up or spin down spinors $|\xi_{sm_s}\rangle$. The ansatz for the ground state is given by a Slater determinant with total orbital momentum $L = l_1 + l_2 = 0$ and total spin $S = s_1 + s_2 = 0$, normally labeled as a spin-singlet state.

Ground state of helium

This ansatz for the ground state is then written as, using the compact notations

$$|\psi_i\rangle = |\phi_{1s}\rangle_i \otimes |\xi\rangle_{s_i m_{s_i}} = |1s, s, m_s\rangle_i,$$

with i being electron 1 or 2, and the tensor product of the two single-electron states as

$|1s, s, m_s\rangle_1 |1s, s, m_s\rangle_2 = |1s, s, m_s\rangle_1 \otimes |1s, s, m_s\rangle_2$, we arrive at

$$\Psi(\mathbf{r}_1, \mathbf{r}_2; s_1, s_2) = \frac{1}{\sqrt{2}} [|1s, 1/2, 1/2\rangle_1 |1s, 1/2, -1/2\rangle_2 - |1s, 1/2, -1/2\rangle_1 |1s, 1/2, 1/2\rangle_2]$$

This is also an example of a state which cannot be written out as a pure state. We call this for an entangled state as well.

Maximally entangled

A so-called maximally entangled state for a bipartite system has equal probability amplitudes

$$|\Psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle.$$

We call a bipartite state composed of systems A and B (these systems can be single-particle systems, or single-qubit systems representing low-lying states of complicated many-body systems) for separable if its density matrix ρ_{AB} can be written out as the tensor product of the individual density matrices ρ_A and ρ_B , that is we have for a given probability distribution p_i

$$\rho_{AB} = \sum_i p_i \rho_A(i) \otimes \rho_B(i).$$

1: Traces of operators

Prove that the trace is cyclic, that is for three operators **A**, **B** and **C**, we have

$$\mathrm{Tr}\{\mathbf{ABC}\} = \mathrm{Tr}\{\mathbf{CAB}\} = \mathrm{Tr}\{\mathbf{BCA}\}.$$

2: Exponentiated operators

Let $\mathbf{A}$ be an operator on a vector space satisfying $\mathbf{A}^2 = 1$ and α any real constant. Show that

$$\exp i\alpha \mathbf{A} = \sum_{n=0}^{\infty} \frac{(i\alpha)^n}{n!} \mathbf{A}^n = I \cos \alpha + i\mathbf{A} \sin \alpha.$$

Does this apply to the Pauli matrices?

3: Reduced density operators I

For each of the Bell states, find the reduced density operator/matrix for each qubit.

4: Reduced density operators II

Suppose we have a composite system which consists of systems A and B in the state $|a\rangle \otimes |b\rangle$, where $|a\rangle$ is a pure state of system A and $|b\rangle$ is a pure state of system B . Show that the reduced density operator of system A alone is a pure state. What about system B ?

The next lecture, February 4, 2026

In our next lecture, we will discuss

1. Discussion of entropy and entanglement
2. Gates and circuits and how to perform operations on states

Reading: Chapters 2.1-2.11 of Hundt's text